
Corrigé — Exercice 33

1)a) $\det(A) = 1 \times 4 - (-1) \times 2 = 6 \neq 0$: A est inversible.

b) $A^2 = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} -1 & -5 \\ 10 & 14 \end{pmatrix}.$

c) $5A - 6I = \begin{pmatrix} 5 & -5 \\ 10 & 20 \end{pmatrix} - \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} = \begin{pmatrix} -1 & -5 \\ 10 & 14 \end{pmatrix} = A^2.$

2)a) De $A^2 = 5A - 6I$ on tire $A(5I - A) = 6I$, d'où $A \times \frac{1}{6}(5I - A) = I$.

b) $A^{-1} = \frac{1}{6}(5I - A) = \frac{1}{6} \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}.$

3)a) $AX = C$ avec $C = \begin{pmatrix} 1 \\ 10 \end{pmatrix}.$

b) $X = A^{-1}C = \frac{1}{6} \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 10 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 14 \\ 8 \end{pmatrix} = \begin{pmatrix} 7/3 \\ 4/3 \end{pmatrix}.$

4) $A^3 = A \times A^2 = A(5A - 6I) = 5A^2 - 6A = 5(5A - 6I) - 6A = 19A - 30I$; puis $A^4 = A \times A^3 = 19A^2 - 30A = 19(5A - 6I) - 30A = 65A - 114I.$